

Observational and model literature for the EXHALE paper planets: $H\alpha$, $Ly\alpha$, He I 10830, and metal transmission lines

This survey gathers, for HD 209458 b, HD 189733 b, WASP-52 b, and WASP-121 b, the observational detections and the theoretical/model studies of the upper-atmosphere transmission tracers targeted by EXHALE: hydrogen $H\alpha$ and $Ly\alpha$, the metastable He I 10830 triplet, and the metal excess-absorption lines. For each model study we record the code or method, the initial conditions/setup, and the reported results. References are given as clickable ADS bibcodes; entries are drawn from an ADS query and its abstracts.

1. HD 209458 b

HD 209458 b (“Osiris”) is the archetypal evaporating hot Jupiter and the first exoplanet with a detected extended, escaping upper atmosphere. Its upper-atmosphere / escape diagnostics have been probed in the hydrogen Lyman- α and H- α lines, the metastable He I triplet at 10830 Å, and a set of metal resonance lines (O I, C II, Si III, Mg I/Mg II, Na I). Below, observational detections and upper limits are collected first, followed by the theoretical / model studies. All entries are drawn from records retrieved through the NASA ADS API; bibcodes are given in typewriter font.

1.1 Observations

Line(s)	Instrument / facility	Result	Reference
$Ly\alpha$ (H I)	HST/STIS	First detection of an extended H atmosphere; $15 \pm 4\%$ absorption in the stellar $Ly\alpha$ core; absorption requires material beyond the Roche limit (escaping H).	Vidal-Madjar 2003, 2003Natur.422..143V
$Ly\alpha$ (H I)	HST/STIS	Data-reduction description of the 2003 $Ly\alpha$ transit ($15 \pm 4\%$); background and airglow subtraction.	Désert 2003, 2003astro.ph.12383D
$Ly\alpha$ (H I)	HST/STIS (reanalysis)	Reaffirms evaporation: $15 \pm 4\%$ over line core (-130 to $+100$ km s $^{-1}$), consistent with $8.9 \pm 2.1\%$ over a wider band; H I must exceed escape velocity / lie outside Roche lobe.	Vidal-Madjar 2008, 2008ApJ...676L..57V
$Ly\alpha$ (H I)	HST/ACS	Transit depth $8.0 \pm 5.7\%$ over the whole $Ly\alpha$ line (two visits averaged); confirms extended exosphere but does not tighten $\dot{M}$.	Ehrenreich 2008, 2008A&A...483..933E

Line(s)	Instrument / facility	Result	Reference
$Ly\text{-}\alpha$ (H I)	HST FUV (STIS archive)	$8.4\text{--}8.9 \pm 2.0\%$ absorption; symmetric, optically-thick Lorentzian profile; argues against huge / strongly asymmetric absorption.	Ben-Jaffel 2008, 2008ApJ...688.1352B
$Ly\text{-}\alpha$ (H I)	HST archive	$8.9 \pm 2.1\%$ absorption but claims <i>no</i> evaporation signature; reports stellar $Ly\text{-}\alpha$ time variability affecting earlier diagnostics.	Ben-Jaffel 2007, 2007ApJ...671L..61B
O I, C II, H I	HST/STIS (G140L)	First heavy atoms in the escaping atmosphere: O I $13 \pm 4.5\%$, C II $7.5 \pm 3.5\%$, H I $5 \pm 2\%$; no absorption in other UV lines; species carried to / beyond the Roche lobe.	Vidal-Madjar 2004, 2004ApJ...604L..69V
C II, Si III	HST/COS	C II $7.8 \pm 1.3\%$ and Si III $8.2 \pm 1.4\%$ transit absorption (-50 to $+50\text{ km s}^{-1}$); tentative velocity structure; $\dot{M} \sim (8\text{--}40) \times 10^{10}\text{ g s}^{-1}$ (solar C).	Linsky 2010, 2010ApJ...717.1291L
C II, Si III	HST/COS	Deep spectroscopy; C II and Si III both $\sim 8\%$ mean transit attenuation; first firm Si detection attributed to long-term variability.	France 2011, 2011Ap&SS.335...25F
Si III (non-det.)	HST/COS + STIS (G140M) reanalysis	<i>No</i> Si III detection: reported COS Si III/C II “absorption” attributed to stellar emission variability; STIS G140M consistent with zero Si III.	Ballester 2015, 2015ApJ...804..116B
Mg I, Mg II	HST/STIS (NUV)	Mg I absorption $6.2 \pm 2.9\%$ (-62 to -19 km s^{-1}); <i>no</i> Mg II signal; probes thermosphere–exosphere transition; Mg I to $\sim 7.5 R_p$; $\dot{M}_{MgI} \sim 3 \times 10^7\text{ g s}^{-1}$.	Vidal-Madjar 2013, 2013A&A...560A..54V
Na I D	HST/STIS + ground high-res	Na I D core profile probes ~ 14 scale heights (1 mbar – 10^{-9} bar); T–P profile with isothermal layer (1500 K) then thermospheric rise to $\sim 2500\text{ K}$; ionization layer near $3 \times 10^{-5}\text{ bar}$.	Vidal-Madjar 2011, 2011A&A...527A.110V (corrigendum 2011A&A...533C...4V)

Line(s)	Instrument / facility	Result	Reference
Na I D	HET (HRS)	Tentative Na I absorption ($-2.63 \pm 0.81 \times 10^{-4}$ (12 Å band); alkali survey with empirical Monte Carlo error control.	Jensen 2011, 2011ApJ...743..203J
Na I D	HIRES/Keck	Excess in-transit Na D ($-108.8 \pm 25.7 \times 10^{-5}$ (narrow band), $\sim 4.7 \times$ larger than continuum / telluric systematics.	Langland-Shula 2009, 2009ApJ...696.1355L
H- α	HET (HRS)	Band-integrated H- α over wide band consistent with zero, but a statistically significant reflection-symmetric feature is present; interpretation ambiguous (contrast with firm H- α in HD 189733 b).	Jensen 2012, 2012ApJ...751...86J
He I 10830	CARMENES (Calar Alto 3.5 m)	First firm He I triplet detection: $0.91 \pm 0.10\%$ (9σ) at mid-transit; net blueshift $1.8 \pm 1.3 \text{ km s}^{-1}$; possible extended tail; strength correlates with X-ray/EUV irradiation.	Alonso-Floriano 2019, 2019A&A...629A.110A
He I 10830	SPIRou (CFHT)	Tentative He I detection; new $\dot{M}$ / temperature constraints from 1D Parker-wind NLTE models.	Masson 2024, 2024A&A...688A.179M
He I 10830	HET/HPF	Confirms the previous (robust/tentative) He I detection for HD 209458 b within a 16-planet survey.	Orell-Miquel 2026, 2026AJ...171..1940
FUV (O I, C II, H I)	HST FUV emission spectra	Non-thermal line broadening required: O I and (probably) C II preferentially heated ($T_{\text{OI}}/T_B \sim 10$, $T_{\text{CII}}/T_B \sim 5$); H I column diagnostics.	Ben-Jaffel 2010, 2010ApJ...709.1284B

Strongest / cleanest detections by line. Ly- α : robust extended-H detection (Vidal-Madjar 2003; $15 \pm 4\%$), with independent confirmations at $\sim 8\text{--}9\%$ over wider bands. He I 10830: firm 0.91% (9σ) CARMENES detection (Alonso-Floriano 2019). Metals: firm O I and C II (Vidal-Madjar 2004), Mg I (Vidal-Madjar 2013); Si III is *disputed* (claimed by Linsky 2010 / France 2011, refuted by Ballester 2015); Mg II *not* detected. H- α : only an ambiguous, non-conclusive feature exists for HD 209458 b (unlike the firm HD 189733 b detection).

1.2 Model studies

Each entry lists the code / method, the initial conditions / setup, and the key results, as stated in the paper abstract. “Not stated in abstract” marks details the abstract does not give.

- **Yelle 2004** (2004Icar...170...167Y). *Code/method*: 1D aeronomical thermosphere model. *Setup*: extra-solar giant planets at 0.01–0.1 AU; EUV heating; H₂/H/H⁺ + H₃⁺ ion chemistry, molecular diffusion, thermal conduction. *Results*: thermosphere heated > 10,000 K, rapid (adiabatic-cooled) escape; lower thermosphere cooled by H₃⁺; abrupt molecular→atomic transition at the thermosphere base.
- **Tian 2005** (2005ApJ...621.1049T; thesis 2005PhDT.....10T). *Code/method*: time-dependent 1D hydrodynamic escape solver (transonic; 2D energy-deposition rather than single-layer heating). *Setup*: neutral-gas escape, validated against isothermal analytic wind. *Results*: transonic Parker-type outflow in good agreement with the HD 209458 b Ly- α observations.
- **García Muñoz 2007** (2007P&SS...55.1426G). *Code/method*: self-consistent 1D physical + chemical aeronomy / hydrodynamic escape model. *Setup*: H/C/O/N chemistry and ionization in the thermosphere; EUV heating at 0.045 AU. *Results*: predicts thermospheric composition and evaporation rate; supports heavy-species presence at high altitude (foundational HD 209458 b model, highly cited).
- **Murray-Clay 2009** (2009ApJ...693...23M). *Code/method*: 1D hydrodynamic (Parker) wind escape model. *Setup*: realistic photoionization heating/cooling, ionization balance, tidal gravity, confinement by host-star wind pressure. *Results*: escape is a dayside Parker wind; main-sequence $\dot{M} \sim 2 \times 10^{10} \text{ g s}^{-1}$ (up to $\sim 2 \times 10^{12}$ in pre-MS); energy-limited scaling $\dot{M} \propto F_{\text{UV}}^{0.9}$ (steeper radiation/recombination-limited regime at high flux); mass loss cannot significantly alter planet mass.
- **Koskinen 2013 I** (2013Icar...226.1678K). *Code/method*: photochemical–dynamical hydrodynamic escape model of the thermosphere. *Setup*: first model including *all* detected species (H, heavy atoms, ions) to explain their high-altitude presence. *Results*: constrains composition, temperature and density profiles; explains heavy-atom transport to high altitude.
- **Koskinen 2013 II** (2013Icar...226.1695K). *Code/method*: companion interpretation paper (empirical + hydrodynamic model comparison). *Setup*: fits H I 1216, O I 1334, C II 1335, Si III 1206.5 Å transit constraints. *Results*: constrains mean temperature and ionization state; implications for the lower atmosphere.
- **Liang 2003** (2003ApJ...596L.247L). *Code/method*: 1D photochemical (upper-atmosphere chemistry) model. *Setup*: H, C, O, H₂, CO, H₂O, CH₄; $T \sim 1000 \text{ K}$, elevated stellar irradiance. *Results*: atomic-H production dominated by H₂O photolysis + OH + H₂; bulk H ~ 3 orders of magnitude above Jupiter.
- **Lecavelier des Etangs 2004** (2004A&A...418L...1L). *Code/method*: analytic energy-balance escape estimate. *Setup*: tidal forces + high thermospheric temperatures; Roche-lobe geometry. *Results*: escape rate matches the Ly- α observation; identifies a new intermediate-temperature “geometrical” escape mode where the exobase reaches the Roche lobe; estimates planet lifetimes.

- **Jaritz 2005** (2005A&A...439..771J). *Code/method*: Roche-potential blow-off analysis. *Setup*: short-period giants incl. HD 209458 b; compares blow-off radius vs. L1 distance. *Results*: HD 209458 b in classical hydrodynamic blow-off (blow-off radius inside L1); distinguishes classical vs. geometrical (Roche-limited) mass loss.
- **Schneider 2007** (2007ApJ...671L...57S). *Code/method*: 3D hydrodynamic (star–planet wind interaction) simulation. *Setup*: planetary wind launched at escape velocity (60 km s^{−1}) into the stellar wind; four runs, $\dot{M}_p = 1.5 \times 10^{-17}$ to $1.5 \times 10^{-14} M_\odot \text{ yr}^{-1}$. *Results*: Ly- α absorption fit gives $\dot{M}_p = (1.8 \pm 0.4) \times 10^{-16} M_\odot \text{ yr}^{-1}$ ($\sim 10^{10} \text{ g s}^{-1}$), consistent with Vidal-Madjar lower limit.
- **Bourrier 2013** (2013A&A...557A.124B). *Code/method*: 3D particle (Monte-Carlo dynamics) model of escaping neutral H. *Setup*: radiation-pressure acceleration + stellar-wind charge exchange; ionizing flux and $\dot{M}$ as free parameters. *Results*: HD 209458 b blue-wing velocities to -130 km s^{-1} explained by radiation pressure alone; ionizing flux $\sim 3\text{--}4 \times$ solar, $\dot{M} \sim 10^9\text{--}10^{11} \text{ g s}^{-1}$; escaping gas forms a cometary tail.
- **Bourrier 2014** (2014A&A...565A.105B). *Code/method*: 3D particle model of neutral + ionized Mg, coupled to an analytic sub-exobase atmosphere. *Setup*: fits the Mg I blue-wing transit absorption; radiation pressure, photoionization, electron recombination. *Results*: $\dot{M}_{\text{Mg}^0} = 2.9^{+0.5}_{-0.9} \times 10^7 \text{ g s}^{-1}$, exobase near Roche lobe ($R_{\text{exo}} \sim 3 R_p$), wind $\sim 25 \text{ km s}^{-1}$; -60 km s^{-1} velocities from radiation pressure; mean sub-exobase $T > 6100 \text{ K}$.
- **Christie 2013** (2013ApJ...772..144C; conf. 2014IAUS...299..281C). *Code/method*: model for H($n=2$) population + H- α line profile. *Setup*: hydrostatic atmosphere in thermal + photoionization equilibrium; collisional and radiative $n=2$ transitions. *Results*: H- α set by an optically-thin $\tau \sim 1$ shell of metastable 2s H below the ionization layer; reproduces HD 189733 b Ly- α +H- α with $10\times$ enhanced ionization, but *cannot* reproduce the asymmetric HD 209458 b H- α profile (model gives symmetric line).
- **Lampón 2020** (2020A&A...636A..13L). *Code/method*: 1D spherical hydrodynamic thermosphere + NLTE He I triplet model + high-res radiative transfer. *Setup*: fits the He I 10830 absorption; H/He ratio, temperature and $\dot{M}$ retrieved. *Results*: H/H⁺ transition at $1.2\text{--}1.9 R_p$; He-triplet peak at $1.04\text{--}1.60 R_p$; H/He $\geq 90/10$ (likely $\sim 98/2$); $\dot{M} = (0.42\text{--}1.00) \times 10^{11} \text{ g s}^{-1}$ at $T = 7125\text{--}8125 \text{ K}$ (energy-limited, $\eta = 0.1\text{--}0.2$).
- **Lampón 2021** (2021A&A...648L...7L; conf. 2021EPSC...15..413L). *Code/method*: 1D hydrodynamic model + He I 10830 and Ly- α analysis. *Setup*: estimates H recombination and advection rates from the line data. *Results*: identifies escape regimes observationally, HD 209458 b is the benchmark *energy-limited* case (HD 189733 b recombination-limited, GJ 3470 b photon-limited).
- **Salz 2016** (2016A&A...586A..75S). *Code/method*: PLUTO-CLOUDY 1D spherically-symmetric radiation-hydrodynamics (photoionization + MHD). *Setup*: 18 strongly-irradiated hot gas planets incl. HD 209458 b; detailed heating/cooling; computes Ly- α absorption/emission. *Results*: reproduces the HD 189733 b Na T–P profile but shows unexplained differences for HD 209458 b; classifies thermospheres by Ly- α absorption (cool winds) vs. emission (hot stable thermospheres); Ly- α cooling / free–free cooling important for compact planets.
- **Salz 2015** (2015A&A...576A..42S). *Code/method*: X-ray-derived high-energy irradiation + energy-limited $\dot{M}$ estimates. *Setup*: Chandra/XMM X-ray observations of nine nearby hot Jupiters incl. HD 209458. *Results*: sev-

eral targets exceed the HD 209458 b mass-loss rate; improved interstellar-absorbed Ly- α luminosities; seven of nine amenable to Ly- α transit spectroscopy.

- **Louden 2017** (2017MNRAS.464.2396L). *Code/method*: differential-emission-measure reconstruction of the stellar XUV spectrum (escape-model input). *Setup*: HST/COS UV line strengths + XMM X-ray, coronal/transition-region model $10^{4.1}$ – 10^8 K. *Results*: H-ionizing luminosity $10^{28.26}$ erg s $^{-1}$ (near mean solar); incompatible with energy-limited $\dot{M}$ from COS unless efficiency $> 40\%$; compatible with early STIS-based $\dot{M}$.
- **Esquivel 2019** (2019MNRAS.487.5788E). *Code/method*: 3D hydrodynamic star–planet wind interaction with radiation pressure + charge exchange. *Setup*: photoionization/recombination/collisional ionization; single stellar-wind parameter set for HD 209458. *Results*: Ly- α absorption dominated by planetary material (+ charge-exchange neutrals); neither radiation pressure nor charge exchange alone reaches ~ 100 km s $^{-1}$, but together with the broad thermal profile they match the observations.
- **Debrecht 2020** (2020MNRAS.493.1292D; thesis 2021PhDT.....10D). *Code/method*: high-resolution 3D hydrodynamic photoevaporation simulation (self-consistent wind launching). *Setup*: ionization + heating from incident high-energy radiation; radiation-pressure term included. *Results*: for the expected Ly- α flux, radiation pressure is *unlikely* to significantly shape the wind or explain the high observed line-of-sight velocities.
- **Hazra 2020** (2020MNRAS.496.4017H). *Code/method*: 1D hydrodynamic escape model driven by a time-varying (solar-cycle) XUV input. *Setup*: observed solar XUV over 1.5 cycles applied to an HD 209458 b-like planet; computes Ly- α and H- α transits. *Results*: $\dot{M}$ varies 7.6 – 18.5×10^{10} g s $^{-1}$ over the cycle; Ly- α variation undetectable, H- α equivalent width varies ~ 1.9 mÅ (potentially detectable); strong flares enhance both lines.
- **Chadney 2017** (2017A&A...608A..75C). *Code/method*: 1D upper-atmosphere thermal-escape model with photo/electron-impact ionization and diffusion. *Setup*: HD 209458 b and HD 189733 b under stellar flares (Sun / ϵ Eri proxies). *Results*: typical flares do not significantly change neutral upper atmosphere or $\dot{M}$; ionospheric electron density rises 2.2 – $3.5\times$ for 3–10 h post-flare.
- **Guo 2010** (2010ApJ...712.1107G). *Code/method*: combined mass-loss + tidal-evolution (analytic) model. *Setup*: energy-limited evaporation with heating efficiency β ; various initial masses/separations. *Results*: evaporation and tidal evolution interact; HD 209458 b physical parameters reproduced.
- **Lilensten 2013** (2013Icar...222..169L). *Code/method*: Boltzmann transport of doubly-charged ions (dications) + thermal photo-ions. *Setup*: Beer–Lambert escape of fast dissociation products; applied to HD 209458 b among solar-system cases. *Results*: dication + thermal-ion escape *cannot* account for the measured C $^{+}$ escape flux of HD 209458 b (marginal but non-negligible source).
- **Linssen 2022** (2022A&A...667A..54L; corrigendum 2023A&A...671C...3L). *Code/method*: Cloudy photoionization code in an iterative scheme over 1D Parker-wind profiles. *Setup*: radiative heating/cooling + expansion cooling + heat advection; self-consistency test resolves the T – $\dot{M}$ degeneracy. *Results*: framework to constrain thermospheric temperature (and thus $\dot{M}$) from He/UV-metal transit spectra; HD 209458 b in the analyzed grid.

- **Linssen 2024 (sunbather)** (2024A&A...688A...43L). *Code/method*: open-source sunbather coupling the p-winds Parker-wind code with Cloudy. *Setup*: isothermal Parker wind + photoionization; computes any known spectral tracer (He 10830, UV metals). *Results*: community tool for $\dot{M}$ retrieval from He and metal-line transit spectra (framework, general to hot Jupiters incl. HD 209458 b).
- **Oklopčić & Hirata 2018** (2018ApJ...855L...110). *Code/method*: simple 1D framework for the metastable He I 10830 population in escaping atmospheres. *Setup*: balance of the He triplet level under stellar XUV/mid-UV; ISM-transparent tracer (complement to $Ly\alpha$). *Results*: establishes the 10830 Å line as a window into escaping atmospheres, the foundational method later applied to HD 209458 b (e.g. Lampón 2020, Alonso-Floriano 2019).

Codes / methods appearing: 1D self-consistent photochemical–hydrodynamic escape models (Yelle; Tian; García Muñoz; Murray-Clay Parker wind; Koskinen I/II), 3D hydrodynamic and 3D Monte-Carlo particle models (Schneider; Bourrier H and Mg; Esquivel; Debrecht), photoionization-coupled Parker-wind frameworks (Salz PLUTO–CLOUDY; Linssen Cloudy / sunbather with p-winds), NLTE He I triplet models (Lampón; Oklopčić & Hirata), $H(n=2)/H\alpha$ models (Christie), analytic energy-balance / Roche-geometry estimates (Lecavelier; Jaritz; Guo), and XUV-reconstruction inputs (Louden; Salz 2015).

2. HD 189733 b

HD 189733 b is the brightest transiting hot Jupiter and has become the archetypal target for upper-atmosphere and escape diagnostics. Its extended thermosphere has been probed in H I $Ly\alpha$, the Balmer $H\alpha$ line, the metastable He I 10830 Å triplet, and a set of metal resonance lines (Na I D, O I, C II, Si III, N V, and the near-ultraviolet Mg/Fe region). A recurring theme is the strong stellar activity of the K-dwarf host, which contaminates the chromospheric-line diagnostics ($H\alpha$, Ca II, Na I D) and drives night-to-night and epoch-to-epoch variability in the reported signals.

2.1 Observations

Line(s)	Instrument	Measured result	Reference
$Ly\alpha$ (H I); O I, C II limits	HST/ACS	$Ly\alpha$ transit depth $5.05 \pm 0.75\%$ (3.5σ above disk); upper limits O I $\lambda 1305 < 11.1\%$, C II $\lambda 1335 < 5.5\%$; first $Ly\alpha$ escape signature	2010A&A...514A...72L
$Ly\alpha$ (H I)	HST/STIS	Spectrally resolved; <i>temporal variability</i> : no absorption at epoch 1, $14.4 \pm 3.6\%$ at epoch 2 over -230 to -140 km s $^{-1}$; high velocities require stellar-wind interaction	2012A&A...543L...4L
$Ly\alpha$; $\lambda 1206.5$, $\lambda 1240$	Si III, N V HST/STIS	Detailed analysis of the Sept. 2011 $Ly\alpha$ transit (false-positive prob. 3.6%); variations in Si III/N V ascribed to a bow shock or stellar variability	2013A&A...551A...63B

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Line(s)	Instrument	Measured result	Reference
O I (FUV)	HST/COS (archival)	Neutral-oxygen in-transit absorption $\sim 6.4 \pm 1.8\%$; requires super-solar abundance and/or super-thermal broadening; flux variability noted	2013A&A...553A...52B
C II $\lambda 1335.7$, H I Ly α , O I	HST/COS + STIS	Tentative but repeatable C II absorption $5.2 \pm 1.4\%$; H I consistent with prior Ly α ; only marginal O I	2023AJ...166...89D
NUV continuum; Mg II/Mg I, Fe II region	HST/STIS (NUV)	Broad excess absorption at ~ 2350 and $\sim 2600 \text{ \AA}$ (Fe II bands); Mg II core absorption ruled out at $2-4\sigma$; no constraint on Mg I	2023A&A...671A.170C
H α	HET/HRS	First H α exosphere detection: $(-8.72 \pm 1.48) \times 10^{-4}$ over a $16 \text{ \AA}$ band; implies excitation temperature $T_{\text{exc}} \approx 2.6 \times 10^4 \text{ K}$	2012ApJ...751...86J
H α , Ca II H&K, Na I D	(HARPS-derived)	Excess H α $0.84\%/1.21\%$ ($5.2/9.9\sigma$), Na I D 0.41% (6.5σ), Ca II up to 2.14% (9.8σ); argues signal is largely chromospheric/stellar	2016MNRAS.462.1012B
H α , Ca II H&K (Balmer pre-transit)	optical high-res spectra	Pre-transit Balmer signal detected but variable in strength/timing; partly correlated with Ca II H (stellar activity); bow-shock geometry not reproduced	2016AJ...152...20C
H α (Balmer)	HARPS + Keck/HIRES	Seven H α transits over 2006–2015; strong epoch-to-epoch variation, no clean Ca II–H α trend; signal likely from extended atmosphere	2017AJ...153..217C
H α (out-of-transit)	high-res optical	Near-transit H α core changes are uncommon in out-of-transit monitoring, strengthening the case for abnormal near-transit variability	2017AJ...153..185C
H α ; Ca II IRT	TIGRE	<i>Non-detection</i> : $< 6.2 \text{ m\AA}$ pre-transit and $< 5.6 \text{ m\AA}$ in-transit (99% conf.) after activity correction; challenges bow-shock/absorption claims	2018A&A...619A..96K
He I 10830	CARMENES	$0.88 \pm 0.04\%$ mid-transit (net blueshift $-3.5 \pm 0.4 \text{ km s}^{-1}$); compact He atmosphere $\sim 0.2 R_p$, column $\sim 4 \times 10^{12} \text{ cm}^{-2}$; activity pseudo-signals up to 80%	2018A&A...620A..97S
He I 10830	GIARPS (GIANO-B + HARPS-N)	$0.75 \pm 0.03\%$ (25σ) in strongest triplet component over 5 transits; night-to-night variability from stellar activity	2020A&A...639A..49G

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Line(s)	Instrument	Measured result	Reference
He I 10830	SPIRou	Detection confirmed in a homogeneous 11-planet survey; He presence correlates with stellar mass and EUV flux; p-winds interpretation	2023A&A...677A..164A
He I 10830	SPIRou	Detection confirmed among 15 planets; new $\dot{M}$ /temperature constraints via 1D Parker-wind + NLTE models	2024A&A...688A..179M
Na I D	HET/HRS	First ground-based Na detection: $(-67.2 \pm 20.7) \times 10^{-5}$ in a narrow band ($> 3\sigma$); $\sim 3\times$ deeper than HD 209458 b	2008ApJ...673L..87R
Na I D	HST/STIS	Na core absorption $0.09 \pm 0.01\%$ (9σ) in $5 \text{ \AA}$; T–P profile shows upper-atmospheric heating (1280 K wings, 2800 K cores)	2012MNRAS.422.2477H
Na I D	HARPS	Spectrally resolved doublet: contrasts 0.64% (D2)/0.40% (D1), excess $0.32 \pm 0.03\%$ (10σ); $T \approx 2600\text{--}3270 \text{ K}$ at 9800–12700 km	2015A&A...577A..62W
Na I D	HARPS	Confirmed: D2 $0.64 \pm 0.07\%$, D1 $0.53 \pm 0.07\%$; effective photosphere $\sim 1.13 R_p$, net blueshift; telluric-correction method comparison	2021MNRAS.502.4392L
Na I D	HARPS / HARPS-N (SLOppy)	Na doublet recovered as validation of the public SLOppy pipeline; consistent with prior HD 189733 b results	2022A&A...667A..19S
Na I, Li I	ESPRESSO	Na doublet confirmed (with careful POLD/activity correction using ANTARES + EvE); Na and Li signatures extracted	2025A&A...700A..41M
Na I D	HARPS-N (GAPS)	Significant Na absorption on most nights in a homogeneous 10-planet search; atmospheric height follows T_{eq} –gravity trend; variability characterized	2025A&A...693A..316S

Note: several optical/UV chromospheric-line results ($H\alpha$, Ca II, Na I D) carry large stellar-activity systematics; the $H\alpha$ non-detection of 2018A&A...619A..96K directly contradicts parts of the earlier $H\alpha$ /bow-shock literature.

2.2 Model studies

Several observational papers above also carry forward models; those model components are described here together with the dedicated theory papers.

Lecavelier des Etangs et al. 2010: 2010A&A...514A...72L. *Code/method:* numerical simulation of escaping neutral hydrogen driven by stellar Ly α radiation pressure (particle/dynamical model). *Initial conditions:* atomic-hydrogen exosphere launched from the planet; free parameters are the escape rate and the ionizing EUV flux. *Key results:* the Ly α light curve is fit with $\dot{M} \approx 10^9\text{--}10^{11} \text{ g s}^{-1}$ and $F_{\text{EUV}} = 2\text{--}40$ times solar; best fit $\dot{M} = 10^{10} \text{ g s}^{-1}$, $F_{\text{EUV}} = 20\times$ solar.

Bourrier & Lecavelier des Etangs 2013: 2013A&A...557A.124B. *Code/method:* 3D particle Monte-Carlo model of the escaping-atom dynamics, forward-modeling Ly α absorption line profiles for both HD 209458 b and HD 189733 b. *Initial conditions:* neutral H atoms released from the upper atmosphere, subject to stellar radiation pressure, photoionization, and (for HD 189733 b) charge exchange/collisions with stellar-wind protons; stellar wind parameterized by temperature, velocity, density. *Key results:* for HD 189733 b the observed -230 to -140 km s^{-1} velocities require stellar-wind interaction (radiation pressure alone insufficient); constrains stellar wind $T \sim 3 \times 10^4 \text{ K}$, $v = 200 \pm 20 \text{ km s}^{-1}$, $n \sim 10^3\text{--}10^7 \text{ cm}^{-3}$, escape rate $4 \times 10^8\text{--}10^{11} \text{ g s}^{-1}$, ionizing flux 6–23 times solar; reveals an “escape-limited” behavior.

Salz et al. 2016: 2016A&A...586A...75S. *Code/method:* 1D spherically symmetric hydrodynamic simulations with the PLUTO–CLOUDY interface (TPCI), coupling detailed photoionization/plasma physics with an MHD solver; a grid of 18 strongly irradiated gas planets including HD 189733 b. *Initial conditions:* high-energy (XUV) irradiated thermosphere; solar composition; radiative heating/cooling including Ly α and free–free emission resolved self-consistently. *Key results:* reproduces the Na I D-derived T – P profile of HD 189733 b; radiative cooling (Ly α) is important for this planet; compact/massive thermospheres (e.g. HAT-P-2 b) found hydrodynamically stable; predicts planetary Ly α absorption/emission signals.

Lampón et al. 2021a: 2021A&A...647A.129L. *Code/method:* 1D hydrodynamic, spherically symmetric outflow model coupled to a non-LTE model for the He I triplet population; applied to HD 189733 b and GJ 3470 b. *Initial conditions:* isothermal-type transonic wind constrained by the measured He 10830 absorption; H density from Ly α used to further constrain temperature, mass-loss rate, and H/He ratio. *Key results:* HD 189733 b has a compact, hot thermosphere with peak $T = 12400^{+400}_{-300} \text{ K}$, very low mean molecular mass (H/He = $99.2/0.8 \pm 0.1$), and near-full ionization above $1.1 R_p$; small radial outflow velocities (contrasted with the extended GJ 3470 b).

Lampón et al. 2021b: 2021A&A...648L...7L. *Code/method:* same 1D hydrodynamic + He-triplet NLTE framework, used to estimate hydrogen recombination and advection rates from combined He 10830 and Ly α absorption. *Initial conditions:* as in 2021a, for HD 209458 b, HD 189733 b, and GJ 3470 b. *Key results:* identifies the three theoretical escape regimes observationally, HD 209458 b is energy-limited, **HD 189733 b is recombination-limited**, and GJ 3470 b is photon-limited; proposes these as benchmark cases.

Dos Santos et al. 2023: 2023AJ...166...89D. *Code/method:* isothermal Parker-wind forward model of the outflow, used to interpret HST UV absorption of C, O, and H nuclei. *Initial conditions:* outflow temperature $\sim 12100 \text{ K}$, solar C abundance; escaping-nuclei altitudes matched to the observations. *Key results:* outflow is mostly neutral within

$\sim 2R_p$ and singly ionized beyond; C II absorption consistent with $\dot{M} \approx 1.1 \times 10^{11} \text{ g s}^{-1}$; predicted O I depth is only comparable to the measurement uncertainty; $Ly\alpha$ comparison implies interaction with the stellar environment.

He 10830 survey models: 2023A&A...677A.164A (Allart et al. 2023) and 2024A&A...688A.179M (Masson et al. 2024). *Code/method:* the SPIRou He 10830 detections are interpreted with 1D isothermal Parker-wind + NLTE radiative-transfer models (the open-source p-winds code in Allart et al.; two independent 1D Parker/NLTE models in Masson et al.). *Initial conditions:* isothermal transonic wind parameterized by thermosphere temperature and mass-loss rate, with line-of-sight velocity and thermosphere altitude fit to the resolved triplet. *Key results:* confirm the HD 189733 b He detection and provide $\dot{M}$ /temperature constraints; both stress tension with some prior literature values and the difficulty of tying He presence uniquely to $\dot{M}$.

Cubillos et al. 2023: 2023A&A...671A.170C. *Code/method (model comparison):* solar-abundance hydrodynamic upper-atmosphere models compared against the observed NUV excess. *Key results:* such models *cannot* reproduce the amplitude of the $\sim 2350/\sim 2600 \text{ \AA}$ features with Fe II absorption at solar abundance, leaving the NUV excess unexplained.

3. WASP-52 b

WASP-52 b is an inflated, low-density hot Jupiter ($M_p \approx 0.46 M_J$, $R_p \approx 1.27 R_J$, $T_{eq} \approx 1300 \text{ K}$) on a 1.75 d orbit around a moderately active K2 V dwarf, sitting at the upper edge of the Neptune desert with a large Roche-lobe filling fraction. Its upper atmosphere has been probed in the metastable He I 10830 triplet, $H\alpha$, and the Na I D / K I resonance lines. No $Ly\alpha$ transit study of WASP-52 b was found in the literature searched here. A recurring complication for every optical study is stellar activity (spot/faculae crossings and the CLV + RM effect), which can mimic or mask planetary absorption.

3.1 Observations

Line	Instrument / method	Result	Reference
He I 10830	Keck II / NIRSPEC (high-res.)	Detection: excess absorption $3.44 \pm 0.31\%$ (11σ), ≈ 66 scale heights, centered on the planet rest frame ($\Delta v = 0.00 \pm 1.19 \text{ km s}^{-1}$)	2022AJ...164...24K
He I 10830	Palomar / WIRC ultranarrowband photometry (0.635 nm filter)	Non-detection; 95th-percentile upper limit $< 0.47\%$ excess absorption	2020AJ...159...278V

Line	Instrument / method	Result	Reference
He I 10830	Palomar / WIRC He survey (Neptune-desert edge)	Tentative detection; outflow much smaller than photoevaporation predictions	2022AJ...164..234V
He I 10830	JWST / NIRISS-SOSS	Detection of He at 7.3σ with evidence of an escaping tail ($\sim 2.9\sigma$); also H_2O (10.8σ) and K hints (2.5σ)	2025MNRAS.539..422F
He I 10830	HET / HPF (high-res. survey of 16 gas giants)	Confirms the previous WASP-52 b He triplet detection	2026AJ...171..194O
He I 10830	VLT / CRIRES ⁺ + CARMENES (high-res., $\phi \simeq \pm 0.1-0.5$)	No significant material beyond transit; no in-transit blueshift (rules out strong day–night anisotropy); notes prior literature range 1.5–5.5%	2025A&A...702A.238N
$H\alpha$	VLT / ESPRESSO (3 transits)	Detection: line contrast $0.86 \pm 0.13\%$, FWHM $11-22 \text{ km s}^{-1}$, no net wind shift; 2nd non-ultra-hot Jupiter with $H\alpha$ after HD 189733 b	2020A&A...635A.171C
$H\alpha$, Ca, Na	Magellan / MIKE (high-res.)	Non-detection of excess absorption; upper limits $< 2\%$ to $< 8\%$; CLV in stellar $H\alpha$ /Ca noted	2026AAS...24711403W
Na ID	VLT / ESPRESSO	Detection: Na D ₁ $1.09 \pm 0.16\%$, Na D ₂ $1.31 \pm 0.13\%$; K D ₁ $0.46 \pm 0.13\%$; planetary origin confirmed by orbital velocity tracking	2020A&A...635A.171C
Na ID	GTC / OSIRIS (low-res., 522–903 nm)	First Na detection; otherwise flat/cloudy spectrum; positive T gradient $0.88 \pm 0.65 \text{ K km}^{-1}$ (possible upper-atmosphere heating)	2017A&A...600L..11C
Na I	HST / STIS + Spitzer (PanCET)	Hint of Na at 2.3σ ; cloudy atmosphere; no K detected	2018AJ...156..298A
Na I	WHT / ULTRA-CAM (Na-filter photometry)	Flat/cloud-dominated transmission; Rayleigh not dominant; stellar bright-region anomaly	2016MNRAS.463.2922K

Line	Instrument method	/ Result	Reference
Na I	VLT / ESPRESSO + 3D NLTE stellar spectra	Sample study of the CLV + RM bias on Na detection (WASP-52 included); methods focus	2024A&A...692A...43C

Ly α : no dedicated WASP-52 b Ly α transit observation was found (see gaps).

3.2 Model studies

Yan et al. (2022): 2022ApJ...936...177Y. *Code/method:* 1D hydrodynamic escape model coupled to a non-LTE level solver, with a new Monte-Carlo Ly α radiative-transfer simulation (spherical stellar Ly α + spherical planetary atmosphere) to obtain the mean Ly α intensity that sets the H($n=2$) population; He(2³S) also solved. *Initial conditions:* escaping-atmosphere outflow; grids over XUV flux F_{XUV} (fraction of a fiducial value) and X-ray fraction β_m ; varied H/He ratio. *Key results:* simultaneous fit of H α and He 10830 requires a high H/He $\sim 98/2$, $F_{\text{XUV}} \approx 0.5 \times$ fiducial and $\beta_m \approx 0.3$; both species originate in the escaping atmosphere; mass-loss rate $\dot{M} \approx 2.8 \times 10^{11} \text{ g s}^{-1}$.

Sharipov et al. (2023): 2023ARep...67...272S. *Code/method:* 3D hydrodynamic atmosphere code; Ly α photon transport by Monte-Carlo; H($n=2$) volume-density maps used to synthesize H α absorption. *Initial conditions:* several XUV ionizing-flux values considered. *Key results:* H α absorption forms in a layer $\sim 1.5 R_p$ thick; recombination Ly α photons (electron–proton recombination) dominate the excitation of H($n=2$).

Nail et al. (2025): 2025A&A...702A.238N (observation + model). *Code/method:* 3D hydrodynamic simulations coupled to an improved radiative-transfer treatment of the He I 10833 triplet, compared with CRIRES⁺/CARMENES high-res. data. *Initial conditions:* outflow morphology grid spanning stream-like (cold) to tail-like (hot) regimes; large Roche-lobe filling. *Key results:* WASP-52 b lies in an intermediate hydrodynamic-escape-parameter regime between stream and tail morphologies; a faint stream could exist below detection limits; no in-transit blueshift rules out strong day–night anisotropy.

Linssen et al. (2022): 2022A&A...667A...54L. *Code/method:* isothermal Parker-wind density/velocity profiles post-processed with the photoionization code CLOUDY in an iterative, self-consistency scheme (radiative heating/cooling + expansion cooling + heat advection). *Initial conditions:* 1D Parker-wind grid, assumed $T = 4000\text{--}12000 \text{ K}$ and $\dot{M} = 10^8\text{--}10^{11} \text{ g s}^{-1}$, fixed composition, no stellar-wind confinement; self-consistency checked in the He 10830 line-forming region. *Key results:* demonstrated on HD 209458 b ($T \approx 8200 \text{ K}$, $\dot{M} \approx 10^{9.84} \text{ g s}^{-1}$); the same procedure constrains T and $\dot{M}$ for WASP-52 b (part of the applied sample).

Vissapragada et al. (2020, 2022): 2020AJ....159...278V, 2022AJ....164...234V (observation + model). *Code/method:* isothermal Parker-wind / 1D atmospheric-escape models and the hydrodynamic outflow code ATES used to interpret the He 10830 photometry and derive mass-loss rates. *Initial conditions:* grids at $T \approx 7000$ and

12000 K; energy-limited outflow framing for the Neptune-desert-edge population. *Key results:* for WASP-52 b, upper limits $\dot{M} \leq 2.1 \times 10^{-4}$ (2.1×10^{-3}) $M_J \text{ Gyr}^{-1}$ at 7000 (12000) K; the measured/limited outflow is much smaller than the ATES prediction, so photoevaporation is too inefficient to carve the upper Neptune-desert edge.

Off-scope models. JWST-era retrieval/GCM work on WASP-52 b (2025AAS...24531602P, 2026ASTCS...1160078S) targets lower-atmosphere chemistry (H_2O , CO_2), clouds, and morning–evening asymmetry rather than the escaping-outflow lines, and is not detailed here.

3.3 Gaps

No $Ly\alpha$ transit observation of WASP-52 b exists in the searched literature. He I 10830 detections span 1.5–5.5% across studies (with one photometric non-detection), reflecting stellar-activity and epoch dependence rather than an agreed value. Reported and modeled mass-loss rates disagree by more than an order of magnitude ($\sim 10^{11} \text{ g s}^{-1}$ from line fits versus much lower He-inferred limits and sub-ATES outflows), leaving $\dot{M}$ poorly pinned.

4. WASP-121 b

WASP-121 b is an ultra-hot Jupiter orbiting very close to its Roche limit, which makes its upper atmosphere a benchmark for atmospheric escape studies. Its transmission spectrum is exceptionally metal-rich, and several atomic features ($H\alpha$, Ca II, Mg II, Fe II, Ba^+) are observed to extend beyond the transit-equivalent Roche-lobe radius ($R_{eqRL} \simeq 1.3\text{--}1.6R_p$), directly tracing the escaping outflow. All entries below are drawn from real ADS records; bibcodes are given in typewriter font.

4.1 Observations

The escape-tracing detections are dominated by VLT high-resolution spectroscopy (ESPRESSO, UVES, HARPS) at optical wavelengths and HST/STIS in the near-UV; the He I triplet was first mapped by JWST/NIRISS. No dedicated $Ly\alpha$ (1215.67 Å) detection for WASP-121 b appears in ADS, and there is no published ground-based He I 10830 line-detection paper prior to the JWST phase curve: both remain observational gaps (the F-type host star weakens the He signal).

Species / line	Instrument	Measured result	Reference
Mg II	HST/STIS	Mg II resolved at 5.9σ , reaching	Sing et al. 2019
2796/2804, Fe II	NUV (PanCET)	$R_{pl}/R_\star = 0.284 \pm 0.037$ (2796 Å);	2019AJ...158...91S
UV1/UV2		Fe II multiplets to $R_{pl}/R_\star \approx 0.3$; both exceed the transit Roche size ($R_{eqRL}/R_\star = 0.158$), i.e. unbound / escaping refractories	

Species / line	Instrument	Measured result	Reference
$H\alpha$ 6562.8, Na I, Fe I	VLT (archival high-res optical)	New $H\alpha$ detection, transit depth $1.87 \pm 0.11\%$, attributed to an extended (escaping) H atmosphere; Na I doublet confirmed; Fe I via cross-correlation (photospheric)	Cabot et al. 2020 2020MNRAS.494..363C
Fe (atomic, limb)	HARPS (HEARTS III)	Reloaded Rossiter–McLaughlin isolates planetary-limb metal (likely Fe I) absorption; line width $14.3 \pm 1.2 \text{ km s}^{-1}$; blueshift $-5.2 \pm 0.5 \text{ km s}^{-1}$ (day–night winds or thermosphere expansion)	Bourrier et al. 2020 2020A&A...635A.205B
Fe I (blue arm)	VLT/UVES	Neutral Fe detected $> 8\sigma$; upper-atmosphere $T = 3710^{+490}_{-510} \text{ K}$; Fe I found physically separated from exospheric Fe II	Gibson et al. 2020 2020MNRAS.493.2215G
$H\alpha$, Ca II, Na, K, Li, Mg, Fe I/II, Cr I, V I	VLT/ESPRESSO	$H\alpha$ ($\sim 1.44 R_p$) and Ca II ($\sim 2 R_p$) extend beyond the Roche lobe with broadening inconsistent with tidally-locked rotation alone (winds / high-altitude jets); Li detected at 6σ	Borsa et al. 2021 2021A&A...645A..24B
H I, Ca II (exospheric) + 17 atomic species	VLT/UVES	Confirms exospheric H I and Ca II; 5 strong detections (incl. Cr I, V I, Ca I, K I), Fe I at 8.8σ , novel Sc II at 4.2σ	Merritt et al. 2021 2021MNRAS.506.3853M
Fe, Cr, V (relative abundances)	VLT/UVES	Retrieved abundances near-solar; anomalously large Mg attributed to the escaping atmosphere not captured by the (bound) model	Gibson et al. 2022 2022MNRAS.512.4618G
Ba^+ , Sr^+ , Co, Ca^+ (+ Ti ⁺ tent.)	VLT/ESPRESSO	First Ba^+ detection (heaviest species then known); Ca^+ shows a clear blueshifted asymmetric feature reaching beyond the Roche lobe, consistent with atmospheric escape	Azevedo Silva et al. 2022 2022A&A...666L..10A

Species / line	Instrument	Measured result	Reference
$H\alpha$, Fe II, Ca II (exospheric)	VLT/ ESPRESSO (multi-epoch)	Retrieved extended features: $H\alpha$ $1.54 \pm 0.04 R_p$, Fe II $1.17 \pm 0.01 R_p$, Ca II $2.52 \pm 0.34 R_p$ (beyond $R_{eqRL} \sim 1.3 R_p$); consistent across epochs/instruments	Maguire et al. 2023 2023MNRAS...519.1030M
Fe, Ni, C, O (refractory vs volatile)	VLT/CRIRES+ + ESPRESSO	Bounded abundances of refractory Fe, Ni and volatile C, O; atmosphere enriched in volatiles relative to refractories (photospheric composition, formation tracer)	Pelletier et al. 2025 2025AJ....169...10P
Fe, CO, V, H ₂ O	HARPS + NIRPS + CRIRES+	Cross-correlation Fe/CO/V (5.8/5.0/4.7 σ); $\Delta K_p = -15 \pm 3 \text{ km s}^{-1}$ circulation offset (photospheric composition/dynamics)	Vaulato et al. 2025 2025A&A...703A.251V
He I 10830 (metastable)	JWST/NIRISS (phase curve)	Absorption persists $> 17 \text{ h}$ (55% of orbit), leading transit by $> 5 \text{ h}$ and trailing by $> 10 \text{ h}$; optically thick region $\sim 110 R_p$ ($\sim 0.1 \text{ AU}$), resolved into leading and trailing tails; not self-consistently reproduced by current models	Allart et al. 2026 2026ASTCS...1130205A

4.2 Model studies

Huang, Koskinen, Lavvas et al. 2023 (2023ApJ...951...123H). *Code / method:* 1D forward model of the upper atmosphere and hydrodynamic escape, extended to include atomic metal species (Mg, Fe, Ca), coupled to a $Ly\alpha$ Monte-Carlo radiative-transfer calculation for the excited ($n=2$) hydrogen population and its heating/ionization feedback. *Initial conditions / setup:* 1D radial escape model driven by stellar XUV; explicitly treats Roche-lobe overflow and requires careful coupling to the structure of the lower and middle atmosphere (base boundary); metal species carried in the ionization/thermal balance. *Key results:* Reproduces the observed Mg, Fe, Ca and $H\alpha$ features extending outside the Roche lobe; line broadening from the Roche-lobe-overflow-driven outflow is required to match observed line widths; the high mass-loss rate is set by Roche-lobe overflow; metals and excited-state H significantly affect the thermal and ionization balance of the thermosphere. (The most directly relevant benchmark for EXHALE-type modeling.)

Yan, Guo, Huang & Lavvas 2021 (2021ApJ...907L...47Y): BOTH (model + data comparison). *Code / method:* XUV-driven 1D hydrodynamic escape simulation coupled to detailed hydrogen level-population and radiative-transfer calculations for $H\alpha$. *Initial conditions / setup:* Fiducial XUV irradiation with a scaling of $0.5\text{--}1.5\times$ to probe stellar-activity variation; 1D transonic planetary wind. *Key results:* Mass-loss rate $\dot{M} \sim 1.28 \times 10^{12} \text{ g s}^{-1}$; high temperature and $Ly\alpha$ intensity populate the H first excited state so the modeled $H\alpha$ transmission matches observations; the supersonic (transonic) wind region contributes most of the $H\alpha$ equivalent width, making $H\alpha$ a good tracer of the escaping atmosphere.

Wang, Lin, Wang et al. 2026 (2026arXiv260201364W). *Code / method:* 3D simulations from the planetary surface to extended outflows using the GPU-accelerated KRATOS framework, coupling hydrodynamics with non-equilibrium thermochemistry and ray-tracing radiative transfer. *Initial conditions / setup:* Full 3D domain including day–night circulation of the lower atmosphere; transonic photoevaporative outflow at altitude; parametric variation of FUV, EUV and X-ray irradiation; optional stellar-wind confinement. *Key results:* Outflow is shaped by stellar gravity and orbital motion into two spiral arms with line-of-sight velocities $\sim 40 \text{ km s}^{-1}$, explaining high-velocity Na and $H\alpha$ absorption without a super-rotation jet; different species trace different regions, Fe (inner, rotation-dominated), Na (dense spiral arms), $H\alpha$ and He 10830 (extended, ionized gas); stellar-wind confinement enhances metastable-He absorption.

Lothringer, Fu, Sing et al. 2020 (2020ApJ...898L...14L). *Code / method:* PHOENIX 1D atmosphere models (chemical-equilibrium, *not* an escape model). *Initial conditions / setup:* Grid of equilibrium temperatures 1000–4000 K, with and without condensate rainout. *Key results:* Near-UV/blue absorption shortward of $0.5 \mu\text{m}$ in ultra-hot Jupiters (incl. WASP-121 b) can arise from equilibrium atomic/ionic opacity (Fe, Ti, Ni, Ca, Cr, Mn, SiO) rather than only escaping metals; provides a competing (bound-atmosphere) interpretation of the NUV excess and a rainout diagnostic.

Gebek & Oza 2020 (2020MNRAS.497.5271G). *Code / method:* Non-hydrostatic exosphere / evaporative transmission-spectrum modeling (alkaline Na I, K I curve-of-growth). *Initial conditions / setup:* Three non-hydrostatic density profiles: escaping, exomoon, and torus. *Key results:* Na I equivalent widths of $\sim 1\text{--}10 \text{ m\AA}$ are driven by evaporation rates $\sim 10^3\text{--}10^5 \text{ kg s}^{-1}$; a toroidal (non-hydrostatic) Na configuration is suggested to be consistent with the WASP-121 b Na I data, breaking the atmosphere–exosphere degeneracy via the D2/D1 line ratio.

5. Unique and meaningful EXHALE analyses for each planet

The literature above shows which lines are detected and which codes have been applied to each planet. This section identifies, for each target, the analysis that EXHALE is distinctively placed to contribute: capabilities that the published models for that planet did not combine in one self-consistent calculation. The distinguishing features of EXHALE are: (i) hydrogen, helium, and the trace metals C, N, O, Mg, Ca, Na, Fe solved in a *single* coupled ionization system, so the metal transmission lines and the H/He escape share the same wind, electron density, and charge exchange; (ii) the metastable He I 10830 triplet and the excited $n = 2$ hydrogen population carried self-consistently, giving He 10830 and $H\alpha$ from one model; (iii) He/H and He/metal diffusive separation; (iv) a channel-by-channel heating/cooling budget; (v) an optional lower-atmosphere boundary from VULCAN photochemistry; and (vi) an L_1 Roche-lobe truncation of the line-of-sight integration.

5.1 The LaRT $Ly\alpha$ radiative transfer recomputation of $H\alpha$

This analysis is present in the companion poster but *not yet* in the paper, and it applies to all four planets. EXHALE_transit.py computes $H\alpha$ under a spherically symmetric $Ly\alpha$ field: the $n = 2$ hydrogen population is taken as $n_{2p} = n_{2p}(r)$ with an escape-probability closure for the $Ly\alpha$ mean intensity. Under real (one-sided) stellar illumination the $Ly\alpha$ mean intensity, and therefore the $H(2p)$ population that produces $H\alpha$, is only *cylindrically* symmetric about the star–planet axis, $n_{2p} = n_{2p}(\rho, z)$. A Monte-Carlo $Ly\alpha$ radiative-transfer calculation with LaRT drops the spherical assumption:

1. exhale_to_lart.py exports the converged EXHALE run ($n_{HI}(r)$, $T(r)$, $v_r(r)$, and the incident stellar $Ly\alpha$ line profile) to a LaRT spherical-atmosphere, stellar-illumination model.
2. LaRT returns the two-dimensional $Ly\alpha$ scattering rate $P_\alpha(\rho, z)$ (per atom, per unit source luminosity).
3. tpm_halpha_lart2d.py scales it by the stellar $Ly\alpha$ luminosity, forms $J_{Ly\alpha, eff}(\rho, z) = P_\alpha/B_{12}$, rebuilds $n_{2p}(\rho, z)$ from the Christie et al. (2013) rate equilibrium, integrates the $H\alpha$ optical depth along the transit sight line ($v_{LOS} = v(r)z/r$), and disk-averages over impact parameter.

He 10830, which is set by the spherically symmetric He 2^3S balance, is unchanged; only $H\alpha$ is recomputed. LaRT can additionally be run with an internal, diffuse $Ly\alpha$ volume source (recombination emission), whose radial scattering-rate profile $P_\alpha^{insitu}(r)$ adds the in-situ $Ly\alpha$ pumping that the escape-probability closure only approximates. Both the stellar-illumination and in-situ LaRT runs already exist for all four planets (lart_runs/bm_* and lart_runs/insitu_*), and the poster demonstrated for WASP-52 b that the L_1 -truncated, LaRT-coupled $H\alpha$ reproduces the Chen et al. (2020) depth once the $Ly\alpha$ flux is scaled. Adding this recomputation to the paper gives a geometrically consistent, scattering-resolved $H\alpha$ prediction that no escape-probability treatment provides, and it is the natural EXHALE tool for the $H\alpha$ questions below.

5.2 HD 209458 b: the missing $H\alpha$ and the coupled metal exosphere

The observational record has a firm $Ly\alpha$ detection, a firm He 10830 detection, and firm O I/C II/Mg I metal absorption, but *no* conclusive $H\alpha$ (in contrast to HD 189733 b), and the Christie et al. (2013) model cannot reproduce a strong $H\alpha$ either. The unique EXHALE analysis is therefore twofold. First, the LaRT $Ly\alpha$ recomputation (Section 5.1) predicts the intrinsic $H\alpha$ depth from the same wind that matches the $Ly\alpha$ and He 10830 data, testing whether the weak/absent $H\alpha$ is the expected outcome of HD 209458 b's $Ly\alpha$ field rather than a modeling shortfall. Second, because O, C, and Mg are solved inside the coupled system, EXHALE predicts the O I 1302, C II 1335, and Mg I 2852 excess absorption and their thermosphere-to-exosphere transition simultaneously with the H/He escape: a single self-consistent account of the detected metals, which the published metal studies treated with separate models. The channel-by-channel budget also identifies whether $Ly\alpha$, metal-line, or adiabatic cooling controls this moderately irradiated wind.

5.3 HD 189733 b: disentangling planetary $H\alpha$ from stellar activity

HD 189733 b has the strongest but most *contested* $H\alpha$: a claimed detection (Jensen 2012; Cauley) versus a non-detection (Kohl 2018), with the chromospheric lines ($H\alpha$, Ca II, Na I D) pervasively contaminated by stellar activity. The high-impact EXHALE analysis is the LaRT $Ly\alpha$ recomputation of the *intrinsic planetary* $H\alpha$: a physically grounded prediction of the wind's own $H\alpha$ helps separate the planetary contribution from the stellar activity that the observations cannot cleanly remove. The in-situ diffuse- $Ly\alpha$ run is especially relevant here because HD 189733 b is the benchmark recombination-limited case, where recombination $Ly\alpha$ pumping of $n = 2$ is largest. Coupled to this, EXHALE predicts Na I D, O I, and C II from the same wind for comparison with the long detection chain, and quantifies the recombination-limited energy budget.

5.4 WASP-52 b: the domain/geometry control of He 10830 and the LaRT $H\alpha$

This is where the poster analysis is most developed and should enter the paper. The He 10830 depth reported across studies spans 1.5–5.5% (with one photometric non-detection), and our Phase-1 run over-predicts it. The poster shows that the over-prediction is a *domain/geometry* effect: truncating the line-of-sight integration at the L_1 Roche point (rather than carrying it to $10R_p$) brings He 10830 into agreement, and the LaRT-coupled $H\alpha$ then reproduces the Chen et al. (2020) 0.86% depth once the $Ly\alpha$ flux is scaled. The unique EXHALE contribution is thus a joint He 10830 + $H\alpha$ + Na I D account from one wind, with the He/H ratio and the L_1 truncation as the controlling parameters, that reconciles the scattered He values and predicts $H\alpha$ with the geometrically consistent LaRT transfer rather than the spherical closure.

5.5 WASP-121 b: the escaping metal lines and a differential test of the closest analog

WASP-121 b is the metal-rich, near-Roche-lobe-overflow case: Mg II and Fe II are seen to exospheric radii (Sing 2019), Ca II extends to $\sim 2.5R_p$, and $H\alpha$ reaches beyond the Roche lobe, while the closest published escape model (Huang et al. 2023) already argues for Roche-lobe-overflow-driven outflow. The distinctive EXHALE analysis is the prediction of the *escaping* metal transmission lines (Fe II, Mg II, Ca II) from the coupled-metal wind: a one-dimensional coupled-metal escape account to set against the three-dimensional and photospheric-composition models that dominate the

literature, together with the L_1 -truncated line-of-sight integration required by the observed extended radii. The LaRT $H\alpha$ recomputation supplies the Balmer prediction (Cabot 2020, 1.87%) for this ultra-hot wind, and because Huang et al. (2023) is the closest methodological analog, WASP-121 b is the natural target for a differential code comparison (mass-loss rate, thermal/ionization structure, and line depths) that quantifies where the coupled-metal treatment and the LaRT $Ly\alpha$ transfer change the predicted spectra.